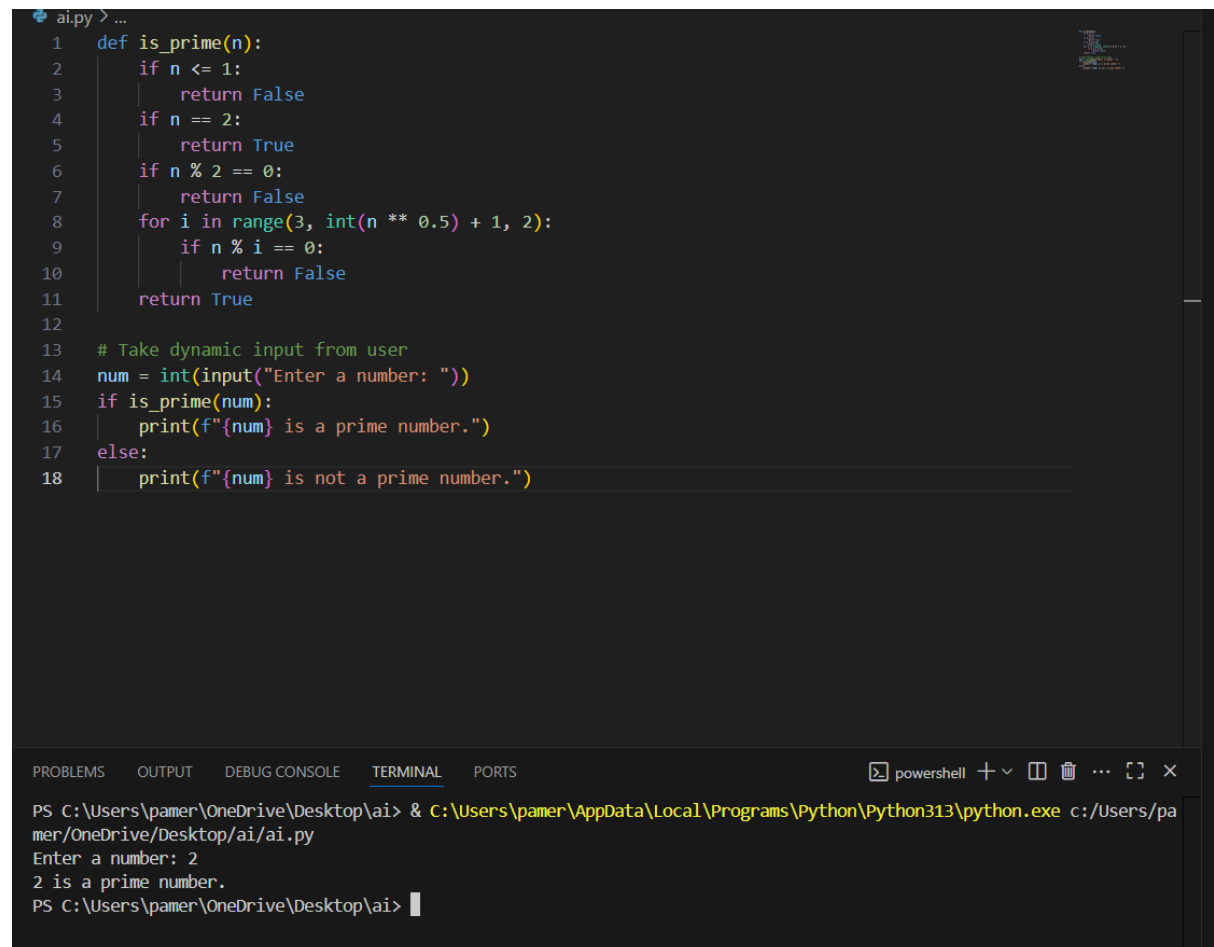


AI-Assisted Coding

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1.Generate a python code to check whether a number is prime or not and take input in dynamic way



```
ai.py > ...
1  def is_prime(n):
2      if n <= 1:
3          return False
4      if n == 2:
5          return True
6      if n % 2 == 0:
7          return False
8      for i in range(3, int(n ** 0.5) + 1, 2):
9          if n % i == 0:
10             return False
11     return True
12
13 # Take dynamic input from user
14 num = int(input("Enter a number: "))
15 if is_prime(num):
16     print(f"{num} is a prime number.")
17 else:
18     print(f"{num} is not a prime number.")

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
powershell + - [] [] ... [] x
PS C:\Users\pamer\OneDrive\Desktop\ai> & C:\Users\pamer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/pamer/OneDrive/Desktop/ai/ai.py
Enter a number: 2
2 is a prime number.
PS C:\Users\pamer\OneDrive\Desktop\ai>
```

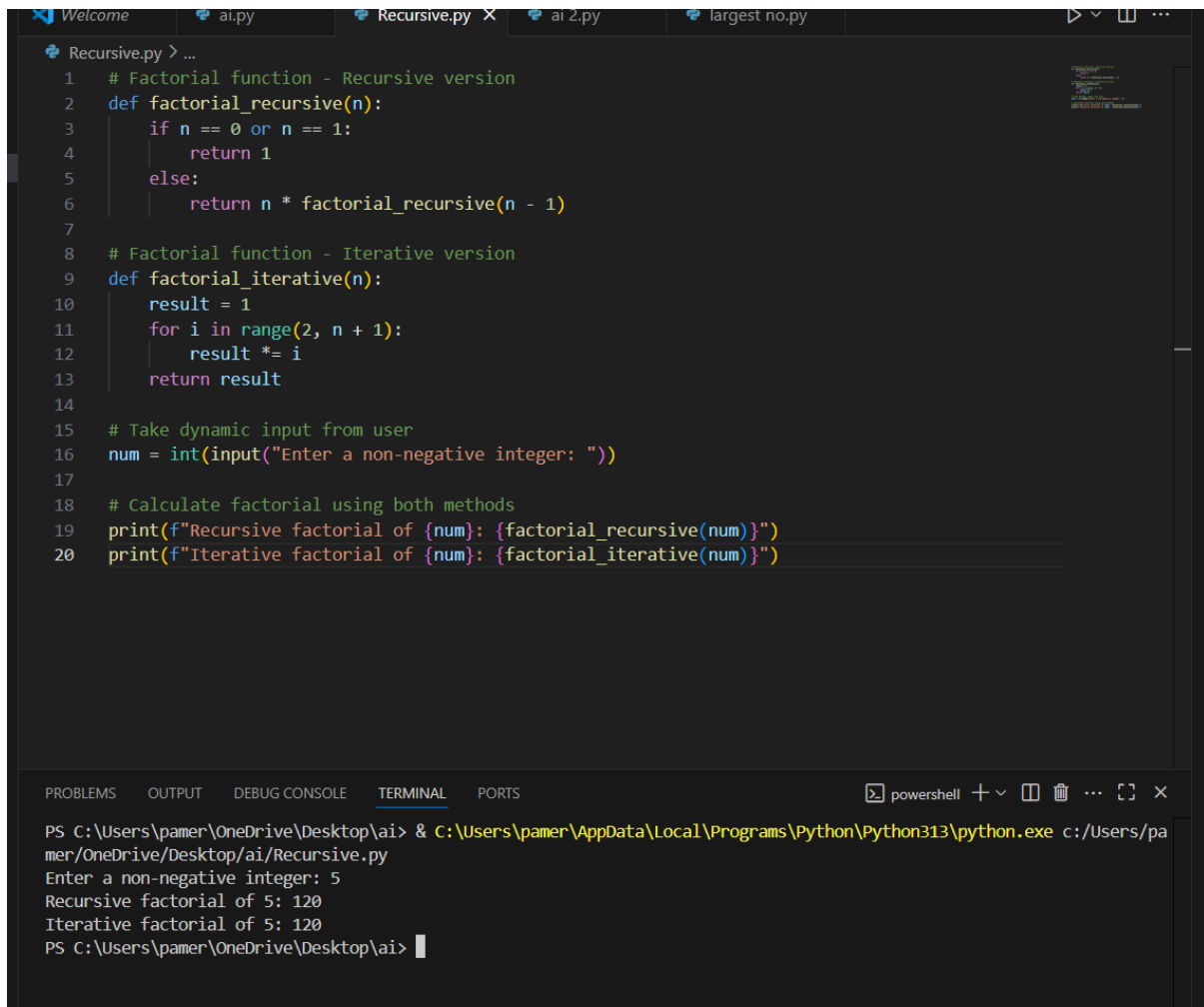
2.Generate a python code to reverse a string using function and take input in dynamic way

The screenshot shows a Visual Studio Code window with a file explorer on the left displaying 'ai_2.py'. The main editor area contains the following Python code:

```
1 def auto_reverse():  
2     |   text = input("Enter text to reverse: ")  
3     |   print("Reversed:", text[::-1])  
4  
5 if __name__ == "__main__":  
6     |   auto_reverse()
```

The bottom panel shows the 'TERMINAL' tab with the command prompt running the script. The user entered 'srikar' at the prompt, and the program printed 'Reversed: rakirs'. The taskbar at the very bottom shows icons for File Explorer, Edge browser, and several instances of PowerShell.

3. Generate both recursive and iterative version of a factorial using comments by python program



```
1 # Factorial function - Recursive version
2 def factorial_recursive(n):
3     if n == 0 or n == 1:
4         return 1
5     else:
6         return n * factorial_recursive(n - 1)
7
8 # Factorial function - Iterative version
9 def factorial_iterative(n):
10     result = 1
11     for i in range(2, n + 1):
12         result *= i
13     return result
14
15 # Take dynamic input from user
16 num = int(input("Enter a non-negative integer: "))
17
18 # Calculate factorial using both methods
19 print(f"Recursive factorial of {num}: {factorial_recursive(num)}")
20 print(f"Iterative factorial of {num}: {factorial_iterative(num)}")
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\pamer\OneDrive\Desktop\ai> & C:\Users\pamer\AppData\Local\Programs\Python\Python313\python.exe c:/Users/pamer/OneDrive/Desktop/ai/Recursive.py
Enter a non-negative integer: 5
Recursive factorial of 5: 120
Iterative factorial of 5: 120
PS C:\Users\pamer\OneDrive\Desktop\ai>

4. Generate a python code to find the largest number in a list and assess code quality and efficiency

```

largest no.py > ...
1  def find_largest(numbers):
2      """
3      The largest number in the list.
4      Raises:
5      ValueError: If the list is empty.
6      """
7      if not numbers:
8          raise ValueError("List is empty.")
9      largest = numbers[0]
10     for num in numbers[1:]:
11         if num > largest:
12             largest = num
13     return largest
14
15 # Code Quality & Efficiency Review:
16 # - The function uses a simple linear scan (O(n)), which is optimal for unsorted lists.
17 # - Handles empty lists with a clear exception.
18 # - Uses clear variable names and docstring for maintainability.
19 # - For very large lists, this is as efficient as possible without parallelization.
20
21 # Dynamic input handling:
22 if __name__ == "__main__":
23     try:
24         user_input = input("Enter numbers separated by spaces: ")
25         numbers = [float(x) for x in user_input.strip().split()]
26         print("Largest number:", find_largest(numbers))
27     except ValueError as e:
28         print("Error:", e)

```

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

powershell

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PS C:\Users\pamer\OneDrive\Desktop\ai> & C:\Users\pamer\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/pamer/OneDrive/Desktop/ai/largest no.py"

Enter numbers separated by spaces: 1 2 3 4 5 6

Largest number: 6.0

PS C:\Users\pamer\OneDrive\Desktop\ai>